

Dosage Compensation in *Drosophila*: Sequence-Specific Initiation and Sequence-Independent Spreading of MSL Complex to the Active Genes on the Male X Chromosome.¹

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Abstract—For the dosage compensation to occur, genes on the single male X chromosomes in *Drosophila* must be selectively bound and acetylated by the ribonucleoprotein complex called MSL complex. It remained unknown how such exquisite specificity is achieved, and whether specific DNA sequences were involved. In the present work we demonstrate that it is transcription of the gene on the X chromosome that is important for MSL targeting, irrespective of gene origin and DNA sequence.

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The MSL complex in *Drosophila* is required for dosage compensation of the male X chromosome and is an excellent system to study the mechanisms by which chromatin regulators recognize their targets and fine-tune transcription levels. MSL consists of at least 5 proteins (MSL1, MSL2, MSL3, MOF, and MLE) and 2 non-coding roX RNAs. It associates with active genes specifically on the male X chromosome where it acetylates histone H4 at lysine 16, and increases expression approximately two-fold [1–5].

One of the central questions in the field of dosage compensation is how the responsible machinery is targeted exclusively to the X chromosome. Our previous work showed that in *D. melanogaster* there exists a set of 150–300 so-called chromatin entry sites (CES) that are enriched on the X chromosome, and which serve as sequence determinants for the initial binding of the MSL complex [6]. The vast majority of these sites share a common consensus GA-rich MRE motif (MSL recognition element). Strikingly, MRE was observed to be both necessary and sufficient to autonomously recruit MSL complex. On the other hand, CES are only *two* fold enriched on the X compared to the autosomes, and conversely, many MRE-like sequences on the X are ignored by the MSL complex. Clearly then, the majority of the ~1000 genes bound

by MSL complex on the male X are unlikely to contain an entry site.

How are such genes recognized by the MSL complex? Historically, binding of MSLs to gene targets on the X was hypothesized to rely on the presence of degenerate DNA sequences at each and every dosage-compensated gene on the X. However, despite many analyses such sequences remained too degenerate to allow identification [7–10].

On the other hand, recent studies have implicated the transcription-dependent H3K36me3 chromatin mark as important for optimal levels of MSL binding in its complete pattern [10–14]. So, one of the possible scenarios would be sequence-independent and transcription-dependent binding of MSL complex to the genes on the X via H3K36me3-assisted spreading from the nearby CES. This model thus predicts that any transcribed and H3K36me3-marked gene will be targeted by the MSL complex in the context of the X chromosome irrespective of its origin and sequence.

To test this prediction, we took sequences that never normally attract the MSL complex, and asked whether they could acquire MSL binding in the context of male X chromosomes simply by being transcriptionally active. We created a construct which we named TrojanHorse, consisting of a 14 kb fragment from chromosome arm 2L encompassing two small genes, *RpL40* and *cg3702* (Fig. 1a). These are

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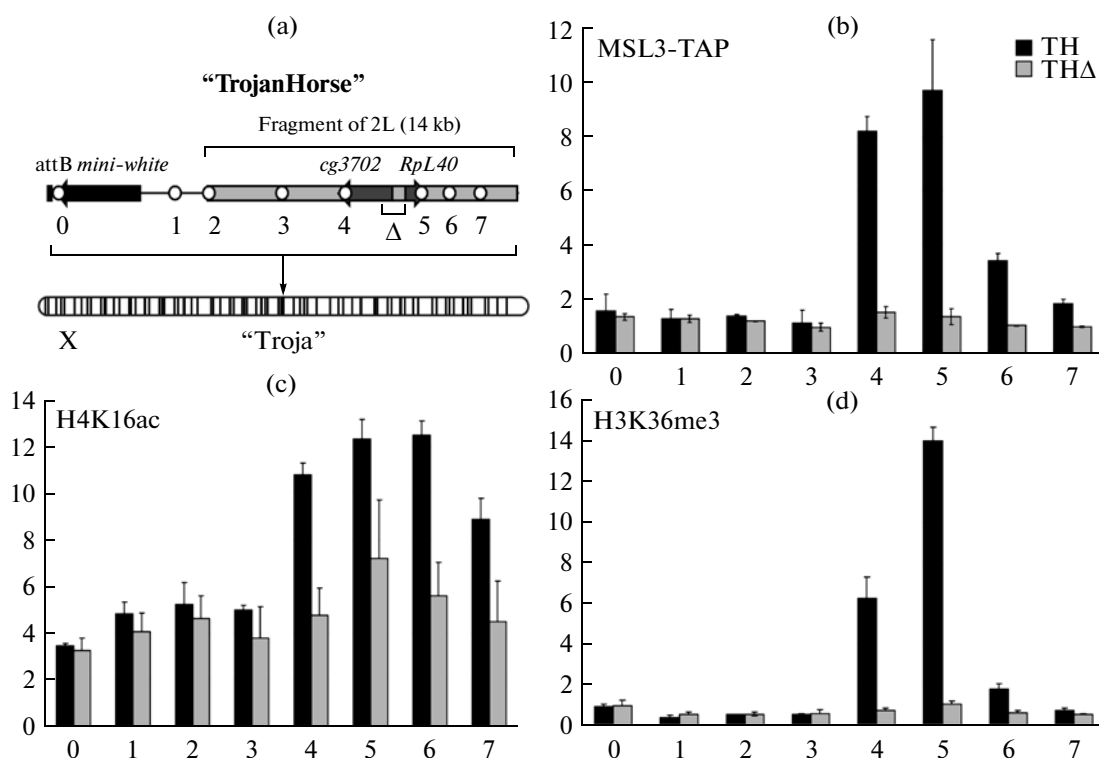


Fig. 1. (a) Structure of the TrojanHorse transposon integrated on the X chromosome. Sequences from the 2L chromosome are shown in gray. Small DNA tags of non-drosophila origin are depicted as white circles. Δ denotes sequences deleted in a "promoterless" version of TrojanHorse. Fold enrichment of MSL3-TAP (b), H4K16ac (c), H3K36me3 (d) across the TrojanHorse (black bars) and TrojanHorse Δ (gray bars), PKA- and input-normalized.

expressed at all stages of development and are associated with H3K36me3, both hallmarks of typical MSL targets on the X. We altered the sequence of the 14 kb fragment by incorporating five 200 kb tags of non-Drosophila origin into the 3'-UTRs of *RpL40* and *cg3702*, and the non-transcribed regions of TrojanHorse. This design allowed us to clearly discriminate between transcripts derived from the endogenous *RpL40* and *cg3702* genes and their transposed copies on the X. Additionally, these tags enabled direct assessment of MSL protein and histone modification profiles across the TrojanHorse insertion on the X chromosome. We placed the TrojanHorse construct in a precise genomic position "Troja" (Fig. 1a) using the recently developed attB/attP phage phiC31 integration system [15, 16]. The insertion site (attP18 at cytological position 6C12) was in a gene-rich region encompassing numerous MSL gene targets.

If specific DNA sequences at each gene are required to complete the MSL pattern on X, then TrojanHorse genes, originating from an autosome, should not acquire MSL binding. Alternatively, if linkage to the X chromosome and the active state are sufficient, TrojanHorse genes should be bound and acquire H4K16ac. Therefore, we performed ChIP assays to determine the binding profiles for TAP-tagged MSL3, within TrojanHorse at Troja (Fig. 1b). We observed

pronounced binding of MSL3 to the tags located at the 3'-ends of both *cg3702* and *RpL40* of TrojanHorse in mixed sex embryos (Fig. 1b) and male larvae (data not shown). In contrast, the untranscribed regions of TrojanHorse did not attract MSL complex. Furthermore, MSL complex binding led to H4K16 acetylation of TrojanHorse chromatin, which peaked at the 3'-UTRs of *cg3702* and *RpL40*, in both embryos (Fig. 1c) and male larvae (data not shown). Finally, we confirmed that the two genes within TrojanHorse were indeed transcribed in larvae and male and female embryos by mapping RNA polymerase II and histone methylation marks expected for active genes: H3K4me2 at the 5' ends, and H3K36me3 biased towards the 3' ends (Fig. 1d and data not shown). Taken together, our results demonstrate that transcribed and H3K36me3-marked autosomal genes can become typical MSL targets when placed on the male X, favoring a sequence-independent model for spreading. Our data also indicate that targeting can occur in the apparent absence of a nearby CES, as the closest identified entry sites are located approximately 54 kb proximal to Troja, indicating that targeting can occur over long distances in cis.

In the context of TrojanHorse, MSL binding correlated with transcription, so we next tested whether transcriptional activity was a prerequisite for the asso-

ciation of transposed autosomal genes with MSL proteins in the male X environment. We created a “promoterless” version of TrojanHorse, where the 1.2 kb region including the promoters and 5'-ends of both *cg3702* and *RpL40*, was deleted (Fig. 1a). This TrojanHorse Δ , otherwise identical to TrojanHorse, was placed in Troja by attP/attB recombination. We confirmed that both genes in TrojanHorse Δ were essentially silent, with 50- and 25-fold decreases in transcript levels for *cg3702* and *RpL40*, respectively (data not shown), and background levels of H3K36me3 (Fig. 1d). Consistent with the loss of transcription and H3K36me3, MSL3-TAP no longer bound non-transcribed *cg3702* and *RpL40* (Fig. 1b), resulting in decreased H4K16ac levels throughout TrojanHorse (Fig. 1c). This direct comparison clearly supports the importance of transcription for MSL recognition of its targets.

To test whether the genes within the intact TrojanHorse were not only associated with MSL complex and marked with H4K16ac, but also were subject to dosage compensation, we directly measured the degree of up-regulation of *cg3702* and *RpL40* in the TrojanHorse context. Indeed, these genes were found to be transcribed and dosage compensated in male larvae so that expression of one copy in males is approximately equal to two copies in females, and about two fold higher than in heterozygous females. Likewise, up-regulation is observed in male embryos.

Taken together, these data indicate that the MSL complex bound to TrojanHorse retains all its functions: it acetylates the underlying chromatin and increases transcriptional output, resulting in dosage compensation of both *RpL40* and *cg3702*. That active autosome-derived genes were bound by MSL complex argues strongly against the idea that each X chromosomal gene possesses special MSL recruiting signals. Instead, we propose that after initial attraction of the MSL complex by a CES or roX RNA gene, transcription serves as the main guiding determinant for MSL complex binding to the majority of active genes on the X. This idea also has an important evolutionary extension, as we imagine that dosage compensation of a newly formed sex chromosome should be easily achieved by mere evolution of recruiting MRE-like sequences with subsequent binding to the active genes, i.e. the targets that need to be compensated, as compared to a less parsimonious scenario, where every gene on the sex chromosome has to evolve a “personal” recruiting signal.

In conclusion, our data allow further refinement of the two-step model for MSL complex recruitment to the male X chromosome. In the first step, MSLs are thought to bind 150–300 chromatin entry sites containing MRE-motifs on X, and ignore autosomes. In the second step, MSLs recognize the active H3K36me3-marked genes on the X irrespectively of their sequence and origin through a transcription-dependent mechanism. Here we speculate on the co-

existence of two modes of MSL spreading on the X: 1) long range spreading from the *roX* genes, and 2) local spreading of MSL complex from the chromatin entry sites scattered throughout the X (with a median distance of about 100 kb). It is only in the context of the X chromosome that both *roX* genes and CES are found *in cis* to each other, making possible both correct chromosome identification and efficient spreading.

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